

# **Intelligenza Artificiale**

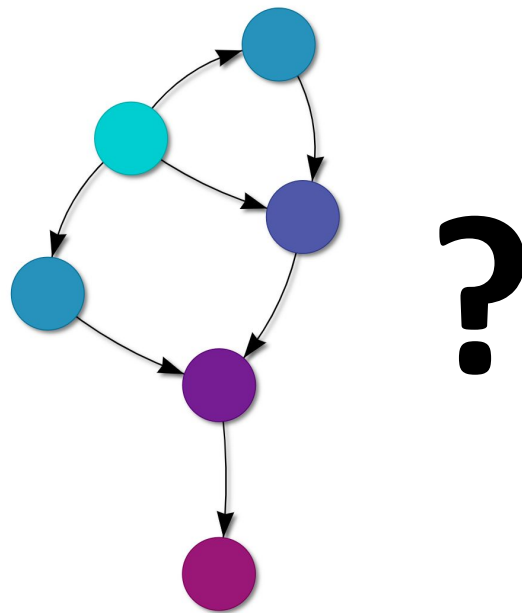
## **“Learning Bayesian Networks”**

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# Lecture outline

- Introduction
- Parameter vs structure learning
- Missing data
- Hidden variables
- Structure learning



Slides partly based on L. E. Sucar instructors material <https://ccc.inaoep.mx/~esucar/Clases-mgp/mgp.html>

# Introduction

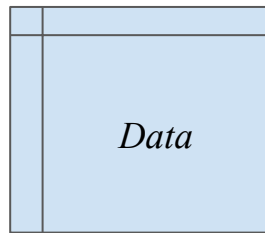
Learning a Bayesian network includes two aspects:

- **Parameter Learning** – When the structure is known, parameter learning consists in estimating the conditional probability tables (CPTs) from data
- **Structure Learning** – There are two main types of methods:
  - global methods based on search and score
  - local methods that use conditional independence tests

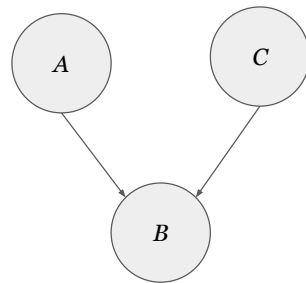
# Parameter learning

- If we have sufficient and complete data for all the variables, and we assume the topology of the BN is known, parameter learning is straightforward
- The CPT for each variable can be estimated from the data based on the frequency of each value – i.e. a **maximum likelihood (ML) estimator**
- For example, to estimate the CPT of  $B$  given it has two parents,  $A$  and  $C$

$$\mathbf{P}(B_i | A_j, C_k) \sim N_{i,j,k} / N_{j,k}$$



| $A$ | $C$ | $P(B A,C)$ |
|-----|-----|------------|
|     |     | ?          |
|     |     | ?          |
|     |     | ?          |
|     |     | ?          |



# Smoothing

- When we estimate probabilities from data, it can sometimes happen that a particular event never occurs, maybe because we don't have enough data
  - the corresponding probability would be zero, implying an impossible case!
- The previous situation can be avoided by using some type of ***smoothing*** for the probabilities, eliminating zero probability values
- There are several smoothing techniques, one of the most common and simplest is **Laplacian smoothing**

# Laplacian smoothing

- Laplacian smoothing consists in initializing the probabilities to a uniform distribution, and then updating these values based on the data
- Consider a discrete variable,  $X$ , with  $k$  possible values: initially, each probability will be set to  $P(x_i) = 1/k$
- If the dataset has  $N$  samples, in which the value  $x_i$  occurs  $m$  times, the estimate of its probability will be

$$P(x_i) = (1 + m)/(k + N)$$

# Laplacian smoothing example

- Let's assume a coin tossed 10 times
  - $k = 2$  (*head* or *tail*) and  $N = 10$
  - if the result is always *head*, i.e.  $m_{head} = 10$  and  $m_{tail} = 0$

$$P(head) = (1 + m_{head})/(k + N) = (1 + 10)/(2 + 10) = 11/12$$

$$P(tail) = (1 + m_{tail})/(k + N) = (1 + 0)/(2 + 10) = 1/12 \neq 0$$



# Incomplete data

Another common situation is **incomplete data**:

- **Missing values** – there are missing values for one or more variables
- **Hidden variables** – a variable or set of variables in the model for which there is no data at all (e.g. *Wind* in the table)

| Outlook  | Temperature 1 | Humidity | Wind | Play |
|----------|---------------|----------|------|------|
| sunny    | xxx           | high     | –    | N    |
| sunny    | high          | high     | –    | N    |
| overcast | high          | high     | –    | P    |
| rainy    | medium        | high     | –    | P    |
| rainy    | low           | normal   | –    | P    |
| rainy    | low           | normal   | –    | N    |
| overcast | low           | normal   | –    | P    |
| sunny    | medium        | high     | –    | N    |
| sunny    | xxx           | normal   | –    | P    |
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| rainy    | medium        | high     | –    | N    |



# Missing values



- There are several alternatives:
  - Eliminate the row with missing values
  - Consider a special “unknown” value
  - Substitute the missing value by the most common value (mode) of the variable
  - **Estimate the missing value based on the values of the other variables in the corresponding row**
- In general the best alternative is the last option

# Estimating missing values

- Learn the initial parameters of the BN based on the complete rows of known variables, then fill-in the missing values and re-estimate the parameters

- In particular, for each row with missing values

1. Instantiate all the known variables in the row
2. Through probabilistic inference, obtain the posterior distribution of the variables with missing values

- e.g. , if *Temperature* is missing, compute  $\mathbf{P}(\text{Temperature} \mid \text{Outlook}=\text{sunny}, \text{Humidity}=\text{high}, \text{Play}=\text{N})$

3. Assign to each missing value in the row the one with highest posterior probability
4. Add this completed row to the database and re-estimate the parameters

| Outlook  | Temperature | Humidity | Wind | Play |
|----------|-------------|----------|------|------|
| sunny    | xxx         | high     | —    | N    |
| sunny    | high        | high     | —    | N    |
| overcast | high        | high     | —    | P    |
| rainy    | medium      | high     | —    | P    |
| rainy    | low         | normal   | —    | P    |
| rainy    | low         | normal   | —    | N    |
| overcast | low         | normal   | —    | P    |
| sunny    | medium      | high     | —    | N    |
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| sunny    | medium      | normal   | —    | P    |
| overcast | medium      | high     | —    | P    |
| overcast | high        | normal   | —    | P    |
| rainy    | medium      | high     | —    | N    |

# Hidden variables

- An approach to estimate the parameters of hidden variables (i.e. their CPTs) is based on the **Expectation–Maximization (EM)** technique
- It consists of two phases which are repeated iteratively:
  - **E step**: the missing data values are estimated based on the current parameters
  - **M step**: the parameters are updated based on the estimated data
- The algorithm starts by initializing the missing parameters with random values

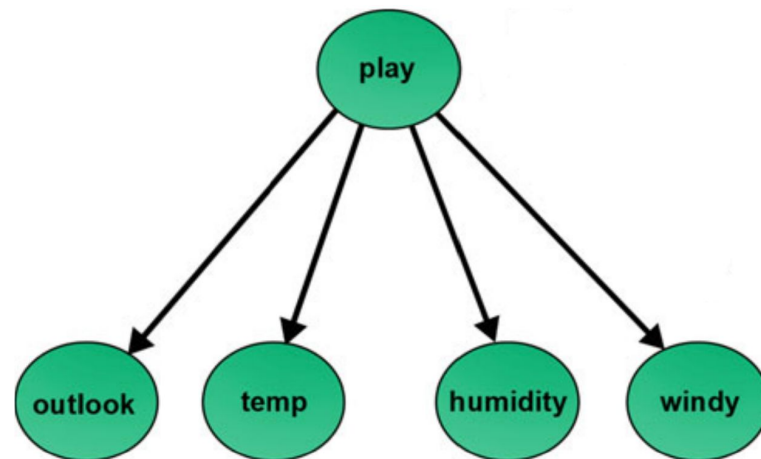
# EM algorithm

1. Obtain the CPTs for all the **complete variables** (those for which the values of the variable and all its parents are in the database) based on a ML estimator
2. Initialize the unknown parameters with random values
3. Considering the actual parameters, estimate the values of the hidden variables based on the known ones via probabilistic inference
4. Use the estimated values for the hidden variables to complete/update the database
5. Re-estimate the parameters for the hidden variables with the updated data
6. Repeat 3–5 until converge (no significant changes in the parameters)

# Example – golf data

| Outlook  | Temperature 1 | Humidity | Wind | Play |
|----------|---------------|----------|------|------|
| sunny    | xxx           | high     | –    | N    |
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- Let's assume the BN is an example of **Naive Bayes Classifier (NBC)**
  - Play* is the *class*, the others are *attributes*



# Example – golf data

- **Missing values** – estimate the probability of temperature for the rows in which it is missing, via probabilistic inference
  - row 1:  $\mathbf{P}(\text{Temperature} \mid \text{sunny}, \text{high}, N)$
  - row 9:  $\mathbf{P}(\text{Temperature} \mid \text{sunny}, \text{normal}, P)$
- **Hidden variable**, *Wind* – we cannot obtain the corresponding CPT,  $\mathbf{P}(\text{Wind} \mid \text{Play})$
- Apply the EM procedure, first with initial random parameters for the CPT:

$$\mathbf{P}(\text{Wind} \mid \text{Play}) = \begin{array}{cc} 0.5 & 0.5 \\ 0.5 & 0.5 \end{array}$$

# Example – golf data

- Given this CPT, we have a complete initial model for the NBC, and can estimate the probability of *Wind* for each row based on the values of the other variables in the same row
- By selecting the highest probability value for each row, we can fill-in the table
- Based on this new data table, we re-estimate the parameters, and obtain a new CPT:

$$\mathbf{P}(\textit{Wind} \mid \textit{Play}) = \begin{matrix} 0.60 & 0.40 \\ 0.44 & 0.56 \end{matrix}$$

- The process is then repeated until the EM procedure has converged

# Example – golf data

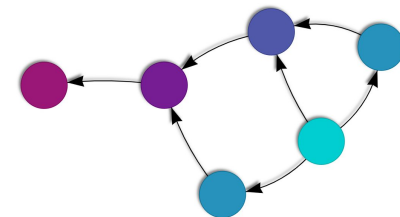
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| Outlook  | Temperature 1 | Humidity | Wind | Play |
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| overcast | high          | normal   | yes  | P    |
| rainy    | medium        | high     | yes  | N    |



# Structure learning



- Structure learning consists in obtaining the topology of the BN from the data
- This is a complex problem because:
  - the number of possible structures is huge even with a few variables (it is super-exponential on the number of variables)
  - a very large dataset is required to obtain good estimates of the statistical measures
- There are several techniques depending on the type of structure – trees, polytrees, **general DAG**
  - We focus on the latter

# General case

For the general case several methods have been proposed, divided into two main classes:

- **Global methods**: perform a heuristic search over the space of network structures, starting from an initial one, and generating a variation of the structure at each step
  - The best structure is selected based on a score that measures how well the model represents the data
  - Common scores are BIC and MDL
- **Local methods**: these are based on evaluating the (in)dependence relations between subsets of variables given the data, to sequentially obtain the structure of the network
  - The most well known variant of this approach is the PC algorithm

covered by future lecture  
on causal discovery

# Global methods

- Global methods search for the best structure based on a global metric
- Different structures are generated and these are evaluated with respect to the data using some scoring method
- There are several variants that depend on two aspects:
  - (i) a **fitness measure (score)** between the structure and the data
  - (ii) a method for **searching** for the best structure

# Scoring functions

- A scoring function evaluates how well a structure fits the data
- Common scoring functions are: the *Bayesian information criterion* (BIC), the *maximum likelihood* (ML), the *Bayesian score* (BD), and the *minimum description length* (MDL) criterion
- The score must balance the precision and complexity of the model
- The **BIC scoring function** includes a penalty term on such complexity

$$\text{BIC} = \log P(D \mid m) - |m| * \log |D| / 2$$

where  $|D|$  is the (constant) number of samples in the data  $D$ , and  $|m|$  is the number of parameters in the model  $m$

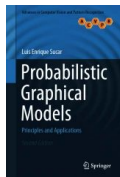
# Search algorithms

- Since the number of possible structures is exponential on the number of variables, heuristic approaches are used for searching for the “best” structure
- One common strategy is to use *hill climbing*:
  1. Generate an initial structure – e.g. a tree
  2. Calculate the score of the initial structure
  3. Add / invert an arc from the current structure
  4. Calculate the score of the new structure
  5. If it improves, keep the change; if not, return to the previous structure
  6. Repeat 3-5 until no further improvements exist
- Setting a **causal ordering** of the variables can reduce the number of potential structures to be evaluated



# Conclusions

- Suggested reading
  - Chapter 8, sec. 8.1-8.2 of L. E. Sucar “Probabilistic Graphical Models: Principles and Applications”



- Next lecture  
Causal inference
- Questions?

6 CFU can skip  
class/workshop on  
**10, 11, 14, 17, 18,  
21 November**

